

Management System Business Process as a Model for the Training of Industrial Enterprises

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Abstract— The development and implementation of innovative solutions in the fields of new production and employing new equipment, as well as such featuring new techniques and technologies, are associated with major expenditure. As the financial resources are limited, the selection of optimum options becomes an issue of crucial importance. It is even more so, as the implementation of high-performance manufacturing technologies, the collection of data on the performance of the units and the processes of the latter, the reorganization of the corporate structure aiming to eliminate the problem-generating system sections, are important to boost the competitiveness of an enterprise, which, in its turn, becomes the main factor of its sustainability and survival on a highly competitive market

Keywords— business processes; versioning; process management

I. INTRODUCTION

The efficiency of implementation of a certain business operation technology or of business restructuring is evaluated based on two types of effects generated by the investment associated with these activities. One of these is comprised of the indicators which characterize the general efficiency of the incremental change, while the other includes the indicators which can be intercompared. The general efficiency indicators cover any increment in the coefficients illustrating the change dynamics of the revenue, production, profit, etc. as compared to the capital investment causing such change. Generally these are calculated to find out what effect the implementation of certain modernization proposal or high-performance manufacturing technology had. However, when the relative indicators and effects are considered, most often they mean the comparison of the reduced expenditures associated with different investment options.

Any changes in the internal structure of an enterprise are associated with the changes in its organizational and manufacturing structure, capital, or property aimed at financial recovery, improved solvency and better performance according to the key indicators in the short term.

During the research and the testing of several proposals on the business processes control system modernization at a helicopter manufacturer, we have found out that the management processes have to be reviewed and updated. The management system has to be able to provide a quicker

reaction to the external factors challenging its efficiency, while also to be able to find an adaptive solution, meeting the modern requirements, as soon as possible. Thus, we have developed an action plan to support the implementation of modern management technology at the enterprise in question. This plan featured the activities aiming at three core fields: business processes control, organizational structure, modernizing business processes control based on versioning.

The implementation of these actions is associated with the following advantages:

- Increased process efficiency due to a shorter production cycle, lower costs, and increased management quality achieved as all of the functions are distributed horizontally. [1]
- The enterprise gains economic motivation to achieve positive and higher financial outcomes in the project. [2]
- Professional follow-up on the timeframe and costs.

When re-designing their structures, any enterprise has specific aims, tasks, and, consequently, requires specific indicators to describe the impact of this process [3]. The system of indicators presented in Table 1 has been developed for the cost-benefit analysis of the enterprise restructuring processes.

We suggest estimating a complex performance indicator based on financial coefficients, i.e. the management performance and efficiency indicators, specific for process-based management. The figures quoted in Tables 1 and 2 are based on the data, provided by the enterprise.

Calculating a complex performance indicator accounting for the financial coefficients of the enterprise is suggested to provide for a complex management performance evaluation. The indicators referred to are used to investigate the performance of the company management system.

The figures used to calculate the performance and impact indicators are presented in the Table 1.

Optimizing the interaction between the processes in place at an enterprise can be done by structuring and normalization of certain business processes and the business in general, as any enterprise is a system the activities of which are aimed at producing positive output. Thus, optimizing it requires a system of indicators.

The hierarchy of the system of indicators, determining the efficiency of an enterprise business process control system

reorganization based on versioning principle, may be presented in the form a tree. (Fig. 1). [5]

TABLE 1. PROCESS-BASED MANAGEMENT PERFORMANCE INDICATORS (ACCORDING TO THE PROCESSES), WITH THE EVIDENCE FROM A HELICOPTER MANUFACTURING ENTERPRISE

Indicator / Deliverable	Calculation formula	Symbol	Actual figure		
			K ₂₀₁₄	K ₂₀₁₅	K ₂₀₁₆
1	2	3	4	5	6
Liquidity(absolute)	C (cash) + S (stocks) / Liabilities (short-term)	K _{ABS.L}	0,96	0,098	0,16
Current liquidity	Assets (current) / Liabilities (short-term)	K _{C.L}	3,1	3,4	3,56
Liquidity(quick)	Short-term AR (accounts receivable) + Short-term FI (financial investments) + C / Short-term liabilities	K _{Q.L}	2,0	2,03	2,25
Turning capability	Capital (CAP) – Assets (noncurrent) / CAP	K _{TC}	0,9	1,0	1,0
Current assets coverage	CAP – Assets (noncurrent) / Assets (current)	K _{CAC}	0,23	0,22	0,18
Autonomy (financial sovereignty, sustainability)	CAP / Balance-sheet total	K _{ABT}	2,1	0,19	0,15
Return on total assets	Profit (balance) / Assets	P _A	0,006	0,002	0,0084
Return on equity	Net profit / CAP	P _E	0,0075	0,0089	0,047
Return on sales	Profit (on sales) / Revenue	P _S	0,1	0,13	0,14
Complex evaluation	$K_{REZ} = \sqrt{(K_{FACT^1} - K_{NORM^1})^2 + (K_{FACT^n} - K_{FACT^n})^2}$	K _{REZ}	2,185	2,194	2,208

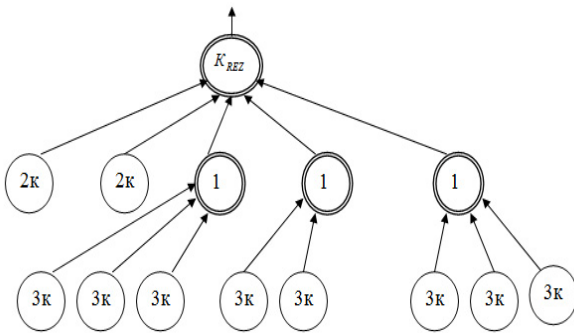


Fig. 1. The hierarchy of the system of indicators, determining the efficiency of an enterprise business process control system reorganization as divided into classes

The top level of the presented tree is formed by the complex enterprise performance indicator K_{RES}, which is based on the intermediate links, i.e. lower-level coefficients, thus establishing the hierarchy of the indicators, describing the efficiency of changes. This allows for the complex monitoring of how effective the decisions made are. Each element has its own class of significance allocated according to the evaluation of its significance and importance in the enterprise profitability generation system. The class and the significance are shown in percentage with the financial standing aspect total equaling 100%. The rules are presented in Table 2.

The data presented in the table has been compiled with due regard to the specific features of the helicopter manufacturing complex, as well as to their relevance. The present research used the indicators characterizing the profitability of a certain enterprise and, consequently, similar guideline values. To make any conclusions on the positive or negative financial

standing of an organization, the indicators should be analyzed over time. Thus, if the evaluation elicits the growth of turnover and profitability indicators, one can conclude that the activity, profit-making capacity, and the performance of the enterprise being analyzed have improved.

The liquidity indicators, which in essence are the financial coefficients and stand for the enterprise's ability to cover its current liabilities with its current assets, have also been studied. [6]

The implementation effect may be expressed with the help of several indicators, but the complex evaluation according to various indicators is based on the main formula, according to which the value is equal to the square rooting of the difference between the norm function and the fact function.

Based on the enterprise performance data for three years (2014–2016), we have calculated the complex performance indicator, the formula of which is as follows:

$$K_{REZ} = \sqrt{(K_{FACT^1} - K_{NORM^1})^2 + (K_{FACT^n} - K_{FACT^n})^2}$$

The need for dynamic response to the issues connected with the enterprise management as well as for the accumulation of the positive experience in taking optimum decisions necessitates monitoring and calculating the economic effect of the improvements made to the business process management based on versioning, which can be expressed in the enterprise financial indicators grouped into four units: liquidity, capital structure, profitability, and turnover coefficients, which when taken in aggregate form the efficiency indicators. [6]

TABLE II. ORIGINAL PERFORMANCE AND EFFICIENCY INDICATORS DATA ON THE HELICOPTER MANUFACTURING ENTERPRISE

Coefficient	Indicator evaluation		
	High	Norm	Low
Absolute liquidity	> 0,1	0,05–0,1	< 0,05
Current liquidity	> 2	1,4–2	< 1,4
Quick liquidity	> 1,0	0,7–1,0	< 0,7
Turning capability	> 0,2	0,1–0,2	< 0,1
Current assets coverage	> 0,5	0,1–0,5	< 0,1
Autonomy (financial sovereignty, sustainability)	> 0,6	0,5–0,6	< 0,5
Return on total assets	> 0,05	0,01–0,5	< 0,001
Return on equity	> 0,15	0,15 and more	< 0,015
Return on sales	> 0,15	0,1–0,15	< 0,15

The suggested method for the performance evaluation of the business-processes management reorganization, which is based on coefficients accounted for during the management of an industrial enterprise, and the original indicators used to forecast the economic effect of the versioning-based management modernization [4], using partial estimates of certain factors, allows for the description of the hierarchy featured in the system of indicators and business processes, determining the effect from the change as well as for numeric calculation.

The resulting complex indicator of management performance at the enterprise under analysis as of 2014 equaled:

$$K_{REZ_{2014}} = \sqrt{(0,096 - 0,05)^2 + (3,1 - 2)^2 + (2,0 - 1)^2 + (0,9 - 0,2)^2 + (0,23 - 0,5)^2 + (2,1 - 0,5)^2 + (0,006 - 0,001)^2 + (0,0075 - 0,15)^2 + (0,1 - 0,15)^2} = 2,185$$

The resulting complex indicator of business processes management performance at the enterprise under analysis as of 2015 equaled:

$$K_{REZ_{2015}} = \sqrt{(0,098 - 0,05)^2 + (3,4 - 2)^2 + (2,03 - 1)^2 + (1 - 0,2)^2 + (0,22 - 0,5)^2 + (0,19 - 0,5)^2 + (0,0016 - 0,001)^2 + (0,0084 - 0,15)^2 + (0,13 - 0,15)^2} = 2,194$$

The resulting complex indicator of business processes management performance at the enterprise under analysis as of 2016 equaled:

$$K_{REZ_{2016}} = \sqrt{(0,16 - 0,05)^2 + (3,56 - 2)^2 + (2,25 - 1)^2 + (1 - 0,2)^2 + (0,18 - 0,5)^2 + (0,15 - 0,5)^2 + (0,009 - 0,001)^2 + (0,06 - 0,15)^2 + (0,14 - 0,15)^2} = 2,208$$

The results obtained prove the effectiveness and positive performance of the versioning-based business-process

management at the enterprise analyzed in 2016, as 2,185 < 2,194 < 2,208.

II. CONCLUSIONS

Thus, based on the difference in the economic performance indicators before and after the implementation of a project, one can evaluate how the project impacted the economic performance of an enterprise. The suggested approach can serve as a basis for a system used to analyze the effect of the business-process management modernization while taking into account the financial activity of an enterprise.

The article suggests a method for the evaluation of business-process management reorganization procedures at the helicopter manufacturing enterprises, based on the complex performance coefficient and the hierarchical system of indicators, illustrating the effect of change. It allows for the complex monitoring of the management decisions effectiveness.

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